# Utilising Useful Commands

So far we've just been understanding the underlying structure for creating scripts. That has **no use** when we have **no idea** *what* commands to use and *how.*

- *So let's begin to understand* ***~5 of the important commands****, because these will make it easier to understand and use other ones.*

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First let's go over **Get-ChildItem**, a command to get "Items" from a folder.

`Get-ChildItem -Path "C:\Folder" (Extra Options)`

- Now these below, **are the *Extra Options.***

-Recurse (Allow to ALSO search the **folders inside target folder**.)

-Filter (Search for patterns in the name, includes wildcards like "*.pdf")

-File (Files only)

-Directory (Folders only)

-Depth # (If there's **folders** inside target folder, *how many folders deep until I stop*?)

-Hidden (Get Hidden Files or not?)

- We've seen this in some of the commands in the previous parts, only now it makes a lot more sense.

Secondly Get-Process (normal program) *[Like on **Task Manager**]*

`Get-Process -Name`

- **Other tags** include:

-ComputerName (Get the process from connected computer)

-Id (ProcessID, might be used for killing a process)

-FileVersionInfo (Metadata)

-IncludeUserName (Who owns this process?)

- Most commonly, *you'd look for the Name with a wildcard (*.exe);* **Then** `Stop-Process Id` **that process.**

Get-Service (background) *[Similar to Processes,* **but usually NOT interacted with.]**

`Get-Service -Name`

-DisplayName (Don't know the technical name, layman term)

-Id (Same as before)

-DependantServices (See what services **require** this one)

-RequiredServices (See what services this one **requires**)

-Include (Pattern matching for wildcards, *print*)

-Exclude (Pattern exclusion for wildcards, *spool*)

- Normally, you'd **use this info** to `Restart-Process -Name spooler -Force`, to *restart the printer service* and all of it's related services.

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All three commands above are normally the FIRST command used. So, here are some commands to use **after piping "|".**

**Firstly,** `Where-Object`

`Get-ChildItem -Path C:\Folder -Recurse | Where-Object { $_.Length -gt 100MB }`

Then, **just do relational operators**; then *the check*

- (> is **-lt**, < is **-gt**, ⩽ is **-ge**, ⩾ is **-le** = is **-eq**, ≠ is **-ne**)

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**Secondly,** `Sort-Object`

`Get-Process | Sort-Object CPU -Descending`

Includes these other **Common options**

-Property (Want to sort multiple, like CPU & RAM)

-Bottom # (**SELECT** bottom 10 from sort)

-Top # (**SELECT** top 10 from sort)

```
> Get-Process | Sort-Object CPU -Descending

 NPM(K)    PM(M)      WS(M)     CPU(s)      Id  SI ProcessName
 ------    -----      -----     ------      --  -- -----------
    141   631.76     349.51   4,152.67    6568   1 Discord
     20    27.91      26.04   2,090.94    4376   0 audiodg
```

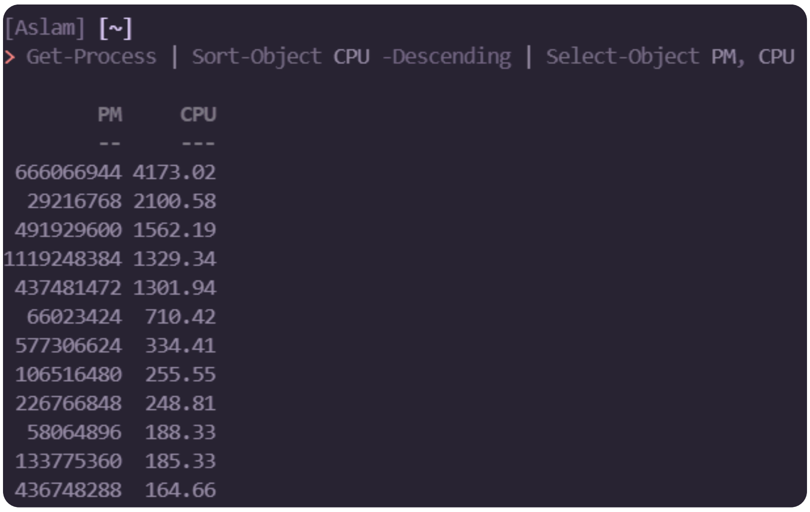
**Thirdly,** `Select-Object`

`Get-Process | Sort-Object CPU -Descending -Top 5 | Select-Object PM, CPU`

Now, both Sort-Object and Select-Object have the ability to pick from the start or end, -Top for Sort, -First for Select. But here we don't use it here, *since Sort already does it.*

- What the **Select does here,** is only spit out certain properties, (Like CPU)

Notice how with Sort, *every property is printed*, **here we get only what we want.**

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Tip -

When **Sorting, Selecting or using Where**; you have to know what *properties the first object has*, and we might <u>not have that memorized</u>.

- *To get that, simply just run your command you wish to filter bare to see the properties on the columns.*


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*These are probably some of the most common Powershell Commands you'll run, That's it for this lab; a different lab will cover the other place you use Powershell.* ***Active Directory & Windows NOS***